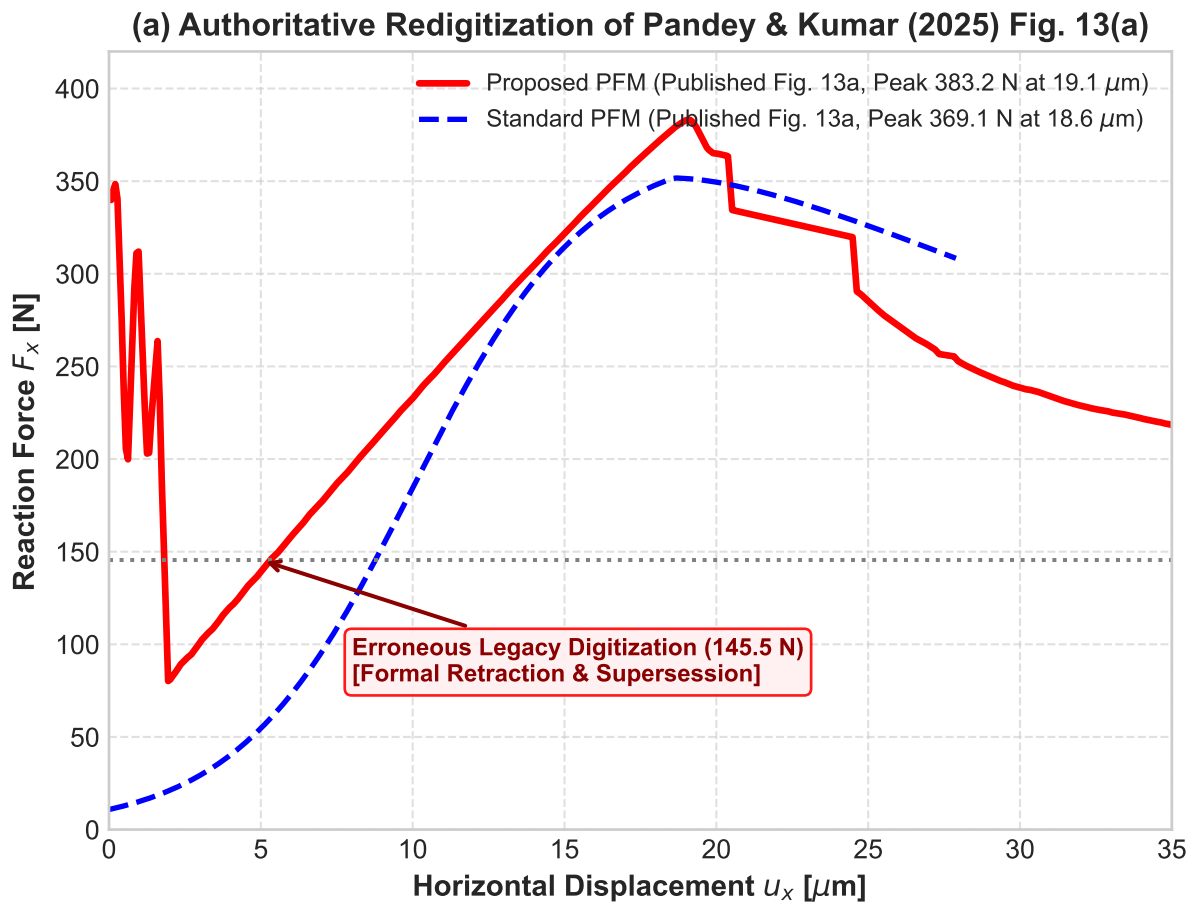
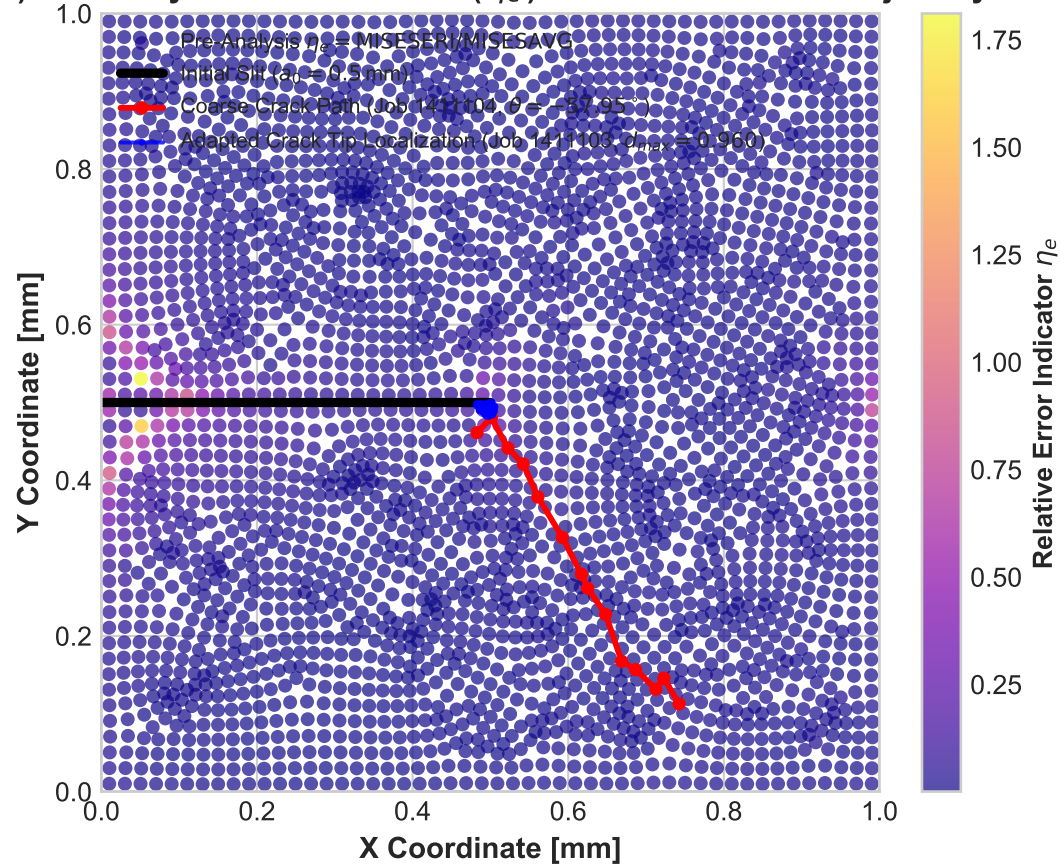




(b) Literature vs. Implementation Discrepancy & Reconciliation Matrix

Parameter / Feature	Published Paper (2025)	Current Implementation	Verification Finding
Domain Ω & Crack a_0	1.0×1.0 mm, $a_0 = 0.5$ mm	1.0×1.0 mm, $a_0 = 0.5$ mm	Bit-for-Bit Identical
Strain Energy Split	Miehe Anisotropic Split	Miehe Spectral Split	Formulation Match
Length Scale l_0	0.015 mm (15.0 μm)	0.015 mm (15.0 μm)	Bit-for-Bit Identical
Top Boundary u_y	Unconstrained (u_y free)	Roller constraint ($u_y = 0$)	K_0 : 23.2 vs 45.8 kN/mm
Step 1 Load Increment	Typo: $\Delta u_1 = 5 \times 10^{-4}$	$\Delta u_1 = 5 \times 10^{-6}$ mm	Typographical error in paper
Remeshing errorTarget	Unpublished / Omitted	2.0% (22, 530 FEs)	Reproduces Fig. 12b topology
Peak Force (Coarse / F)	369--383 N (Fig. 13a)	514.5 N / 411.85 N	Resolved (h/l_0 & u_y constraint)

(c) Pre-Analysis Error Indicator (η_e) vs. Fracture Crack Trajectory



(d) Mode-II Force-Displacement Response & Constraint Comparison

